BioRydberg 2.0

Geometry-Aware Quantum Transport

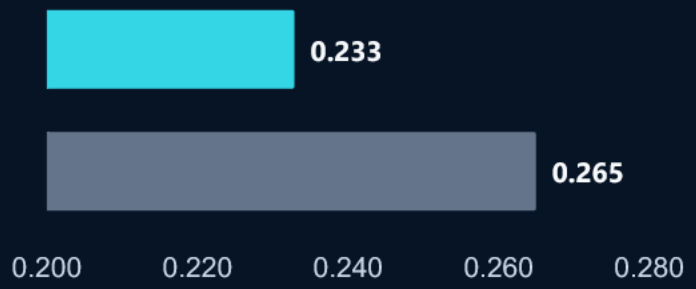
Syed Marjan Ghazi • Pasqal QoolQit Contest 2026

Research question

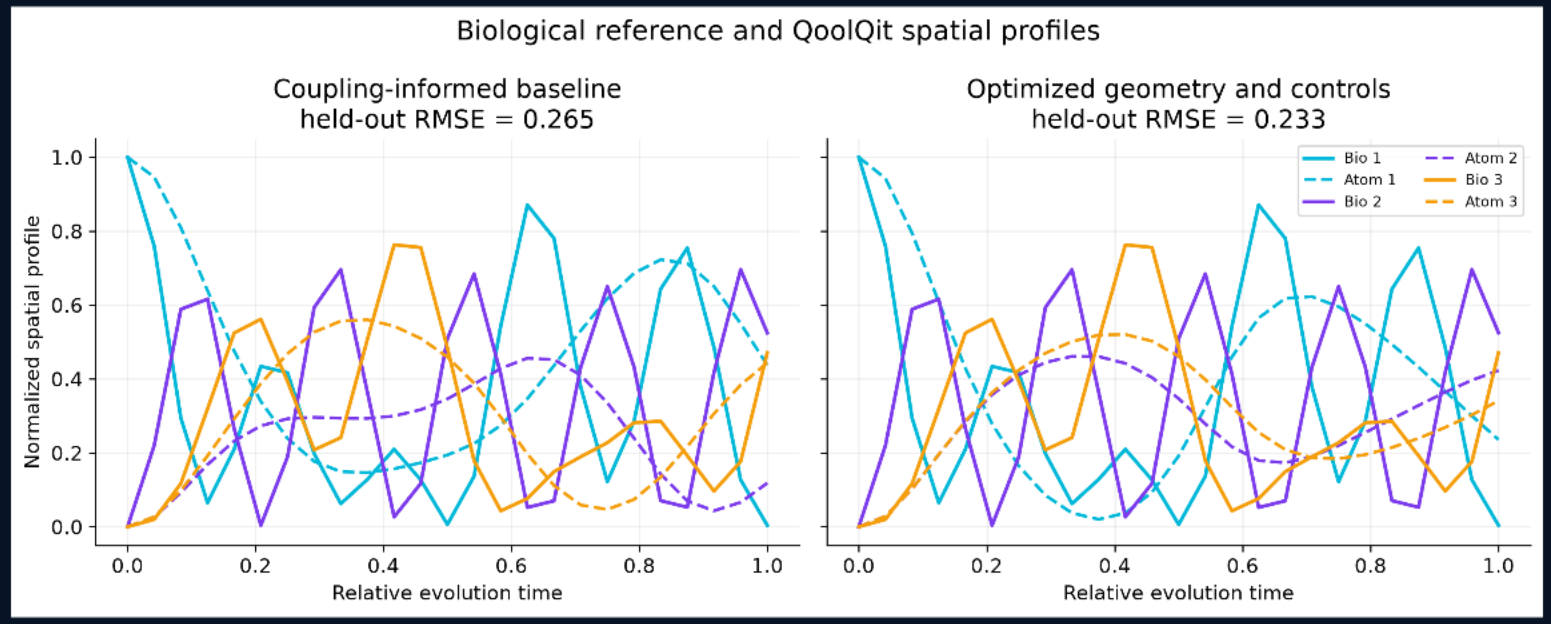
Can bounded atom geometry and laser controls improve a neutral-atom spatial pattern match to a three-site coherent transport reference?

Method

Biological matrix exponential
Coupling-informed 2D register
QoolQit compile + LocalEmulator
Two-seed bounded differential evolution



12.1% lower held-out RMSE



Result: RMSE 0.2650 → 0.2329 on odd time samples excluded from tuning

Interpretation: improved normalized site-distribution similarity only. The biological hopping and driven Rydberg Hamiltonians remain physically distinct.

Experience with QoolQit 1.4.0

Local, credential-free execution on Python 3.12.9

What worked well

Dimensionless model

Register geometry, drive amplitude, detuning, and duration stayed readable before hardware compilation.

Composable execution

Register + Drive + QuantumProgram + AnalogDevice + LocalEmulator formed a clear reproducible path.

Useful observables

Site occupations, stored evaluation times, and the register interaction matrix supported direct validation.

Verification 10 automated tests passed, including a compiled local-emulator run.

Challenges and feedback

Initial-state examples

A localized Rydberg initial state materially changes this use case. More QoolQit examples would make that capability easier to discover.

Sweep ergonomics

Compilation and emulation were fast after import, but batch execution, caching, and progress telemetry would simplify optimization studies.

Result extraction

Using QoolQit's stored result times avoided exact floating-point lookup failures. A documented array-first extraction pattern would help.

Next step Test local detuning controls and noise robustness, then compile the same transparent workflow for Pasqal hardware when access is available.

No cloud or QPU execution claimed